6TH GRADE MATH · UNIT 4: EXPRESSIONS AND EQUATIONS

Arcade Party Planner

Write expressions and solve equations to plan a birthday party at the arcade.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your cousin wants a birthday party at Pixel Palace Arcade. Each guest needs a game card, the party room costs a flat fee, and prizes are bought with tickets.

You will write expressions and solve equations to compare party packages and plan a party that fits the budget.



Party Room

$40 flat fee

up to 3 hours

Game Card

$12 per guest

1 credit = 1 game

Learning Goals

- I can write expressions with a variable for real costs.
- I can evaluate expressions by substituting values.
- I can solve one-step equations to answer questions.

Project Timeline

When	What you will do
Day 1	Write the Costs
Day 2	Use the Properties
Day 3	Solve and Plan
Final	Party Package Proposal + reflection

Part 1: Write the Costs

Day 1 · Let g be the number of guests.

Arcade Price Board

Game Card	Pizza	Drinks
$12 per guest	$15 per pizza	$2 per guest
60 credits	serves 4	

Party Room	Prize Tickets	Cake
$40 flat fee	Ticket bonus: 50	$25 per cake
	per guest	

1. Write an expression for the cost of game cards for g guests.

   Starter: ___ × *g*

2. Write an expression for the cost of game cards and drinks for g guests.

3. Write an expression for the total cost of the room, game cards, and drinks.

4. Evaluate your total-cost expression for 8 guests.

Part 2: Use the Properties

Day 2 · Expressions can be written in more than one way.

5. Show that $12(g + 2)$ and $12g + 24$ are equal when $g = 5$.

6. Pizzas cost $15. Write two equivalent expressions for the cost of p pizzas and 1 cake.

7. A prize costs 2^5 tickets. How many tickets is that?

8. Write an expression for the number of pizzas needed for g guests if each pizza serves 4.

Part 3: Solve and Plan

Day 3 · Write and solve an equation for each question.

9. The game cards cost $132. How many guests came?

Starter: *12g = 132, so g = 132 ÷ ___*

10. Each guest gets 50 bonus tickets. The group has 600 bonus tickets. How many guests are there?

11. Plan a party for 10 guests with game cards, drinks, the room, pizza, and cake. Find the total cost.

Final Product: Party Package Proposal

Your final product must include:

- An expression for each cost
- Your total-cost expression
- The cost for your guest list
- One equation you solved
- A party flyer with prices

Reflection

R1. What does the variable g stand for in your expressions?

R2. How did you check your equation answers?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget. Goes beyond the minimum with a creative, realistic choice.	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Polished, organized, and visually engaging.	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.